Vision Disruptors – Week Progress Report

BraTS 2024 MRI Modality Synthesis | TAMS-WDM Architecture

GROUP NAME: Vision Disruptor
Project: Synthesizing Missing MRI Modalities for Brain Tumor Segmentation
Report Period: Week 7
Date: 11/04/2026

This week, we ran a couple of pilot tests and got pretty decent results. We have evaluated the results on PSIM and SSIM metrics. Below are the image results.

```
======================================================================
  Evaluating 1 patients | contrast: t1c
  Directory : /kaggle/working/results/val_t1c
======================================================================
  Patient                          PSNR (dB)      SSIM           MSE
  ------------------------------  ----------  --------  ------------
  BraTS-GLI-02073-100                 21.611    0.8054    0.02463003

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  Summary [T1C]   N=1
  ──────────────────────────────────────────────────────────────────
  Metric              Mean         Std      Median
  ──────────  ──────────  ──────────  ──────────
  PSNR (dB)         21.611       0.000      21.611
  SSIM              0.8054      0.0000      0.8054
  MSE           0.02463003  0.00000000  0.02463003
  ──────────────────────────────────────────────────────────────────

  Metrics saved → /kaggle/working/results/val_t1c/metrics.csv
         patient_id contrast  PSNR_dB    SSIM     MSE
BraTS-GLI-02073-100      t1c  21.6111 0.80543 0.02463
               MEAN      t1c  21.6111 0.80543 0.02463
                STD      t1c   0.0000 0.00000 0.00000
             MEDIAN      t1c  21.6111 0.80543 0.02463
```

```
======================================================================
  Evaluating 5 patients | contrast: t1c
  Directory : /kaggle/working/results/val_t1c
======================================================================
  Patient                          PSNR (dB)      SSIM          MSE
  ------------------------------  ----------  --------  ------------
  BraTS-GLI-02073-100                 22.254    0.8193    0.02124224
  BraTS-GLI-02073-101                 23.954    0.8410    0.01491259
  BraTS-GLI-02095-100                 24.810    0.8261    0.01311059
  BraTS-GLI-02095-101                 23.595    0.8444    0.01701385
  BraTS-GLI-02095-102                 24.052    0.8315    0.01545712

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  Summary [T1C]   N=5
  ──────────────────────────────────────────────────────────────────
  Metric              Mean         Std      Median
  ──────────────  ──────────  ──────────  ──────────
  PSNR (dB)         23.733       0.839      23.954
  SSIM              0.8325      0.0093      0.8315
  MSE           0.01634728  0.00274707  0.01545712
  ──────────────────────────────────────────────────────────────────

  Metrics saved → /kaggle/working/results/val_t1c/metrics.csv
        patient_id contrast  PSNR_dB     SSIM      MSE
BraTS-GLI-02073-100      t1c  22.2538 0.819340 0.021242
BraTS-GLI-02073-101      t1c  23.9540 0.840950 0.014913
BraTS-GLI-02095-100      t1c  24.8096 0.826124 0.013111
BraTS-GLI-02095-101      t1c  23.5949 0.844408 0.017014
BraTS-GLI-02095-102      t1c  24.0519 0.831503 0.015457
               MEAN      t1c  23.7328 0.832465 0.016347
                STD      t1c   0.8387 0.009253 0.002747
             MEDIAN      t1c  23.9540 0.831503 0.015457
```

Here, the PSIM and SSIM results are still low below the standards, cause the model is still undertrained. The checkpoints used for results generation are 24000 steps. We have also attached the synthesized vs original images in the results section of GitHub. They are blurry, capturing only the overall anatomy, because the system is undertrained.